School Library Management System

1.1 Background of the Organization

A school library is an important part of an educational institution because it supports students and teachers by providing access to books, reference materials, and learning resources. The library manages different types of books such as textbooks, novels, journals, and research materials. It also keeps records of students who borrow and return books, monitors book availability, and handles overdue books and fines. In many schools, these activities are still managed manually using paper records, which make the process slow, difficult, and prone to errors.

A School Library Management System helps organize library operations by storing data electronically, improving efficiency, reducing mistakes, and making it easier to manage books and student records using both MySQL and MongoDB databases.

1.2 Problem Statement

Many school libraries still use manual systems such as notebooks, paper files, and library cards to manage books and student borrowing activities. This creates several problems in daily library operations. Books are tracked on paper, which makes records easy to lose, damage, or duplicate. It becomes difficult for librarians to know which student borrowed a specific book and whether the book has been returned or not.

Another major problem is checking book availability. Since there is no computerized system, students and librarians cannot quickly check if a book is available, already borrowed, or out of stock. This causes delays and confusion during borrowing and returning processes. Managing overdue books and calculating fines also becomes difficult because there is no automatic system for tracking due dates. In addition, preparing reports such as borrowed books reports, overdue lists, and student borrowing history takes a lot of time and effort. These problems reduce efficiency and affect the quality of library services

1.3 Objectives

General Objective

To design and implement an efficient School Library Management System using MySQL and MongoDB that improves the management of books, students, borrowing activities, and report generation.

Specific Objectives

The specific objectives of this project are:

- To create a database for storing book information such as title, author, category, publisher, and availability status
- To maintain accurate student records including student ID, full name, department, year of study, and contact information
- To manage borrowing and returning transactions of books efficiently
- To track overdue books and calculate fines for late returns

- To improve book searching and availability checking
- To generate reports such as borrowed books report, overdue books report, and fine collection report
- To reduce paperwork and minimize data loss caused by manual systems
- To improve the speed, security, and reliability of library services

1.4 Scope of the Project

This project focuses on the main activities of the School Library Management System and covers the essential operations required for effective library management. The system includes book catalog management, where all books are recorded with details such as book title, author name, ISBN, category, publisher, publication year, and availability status. It also manages student records by storing important information like student ID, full name, department, year of study, and contact details.

The project also covers borrow and return tracking by recording which student borrowed a specific book, the borrow date, due date, and return date. This helps the librarian monitor issued books and identify overdue books easily. In addition, the system includes fine calculation for students who return books late based on the school’s library policy. Reports such as borrowed books reports, overdue book reports, and fine payment reports are also included to support better library management and decision-making.

1.5 Benefits of the Project

The School Library Management System provides several important benefits for the school, students, and librarians.

It improves book management by allowing librarians to quickly search, update, and monitor all books in the library. It helps students save time when checking whether a book is available.

The system improves borrowing control by keeping accurate records of issued and returned books. It also helps reduce lost books because every transaction is properly recorded.

Automatic overdue tracking and fine calculation improve accountability and ensure students return books on time.

The database system reduces paperwork, minimizes human errors, improves data security, and makes report generation faster and easier.

1.6 Tools and Methodology

Tools Used

MySQL:- Used for creating and managing the relational database tables such as Books, Students, Borrowing, Librarians, and Fine records**.**

MongoDB:-Used for storing flexible document-based data such as borrowing history, logs, and student activity records.

MySQL Workbench:-Used for writing SQL queries, creating tables, and testing database relationships**.**

MongoDB Compass:-Used for managing collections and testing MongoDB queries.

Draw.io:-Used for drawing ER Diagrams, Logical Schema, and database design structures.

GitHub:-Used for project submission, version control, and collaboration among group members.

Microsoft Word:-Used for writing the final report documentation.

PowerPoint:-Used for preparing project presentation slides.

Methodology Used

The project follows a step-by-step methodology:

Requirement Gathering

Understanding how the school library currently works and identifying major problems.

System Analysis

Studying the current manual system and defining system requirements.

Database Design

Identifying entities, attributes, and relationships and preparing the ER Diagram and Logical Schema.

Normalization

Normalizing the database up to BCNF to remove redundancy and improve consistency.

Implementation

Creating the database using MySQL and MongoDB and inserting sample records.

Testing

Testing queries, relationships, transactions, and report generation.

Documentation

Preparing the final report, presentation slides, and GitHub project submission.

This methodology ensures that the final system is complete, reliable, and suitable for real-world school library management.

Overall, the project increases efficiency, improves service quality, and supports better decision-making for school library management.